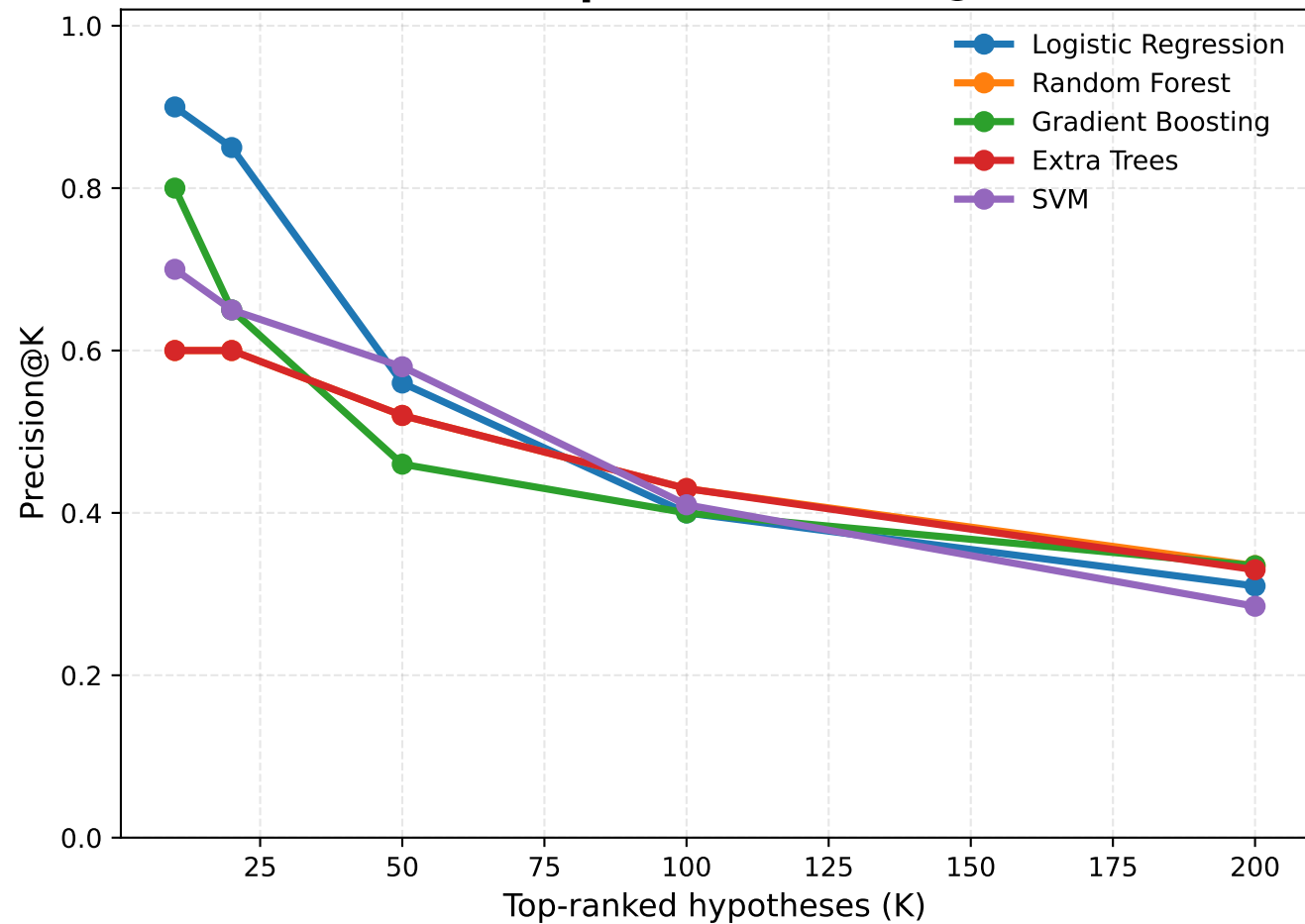
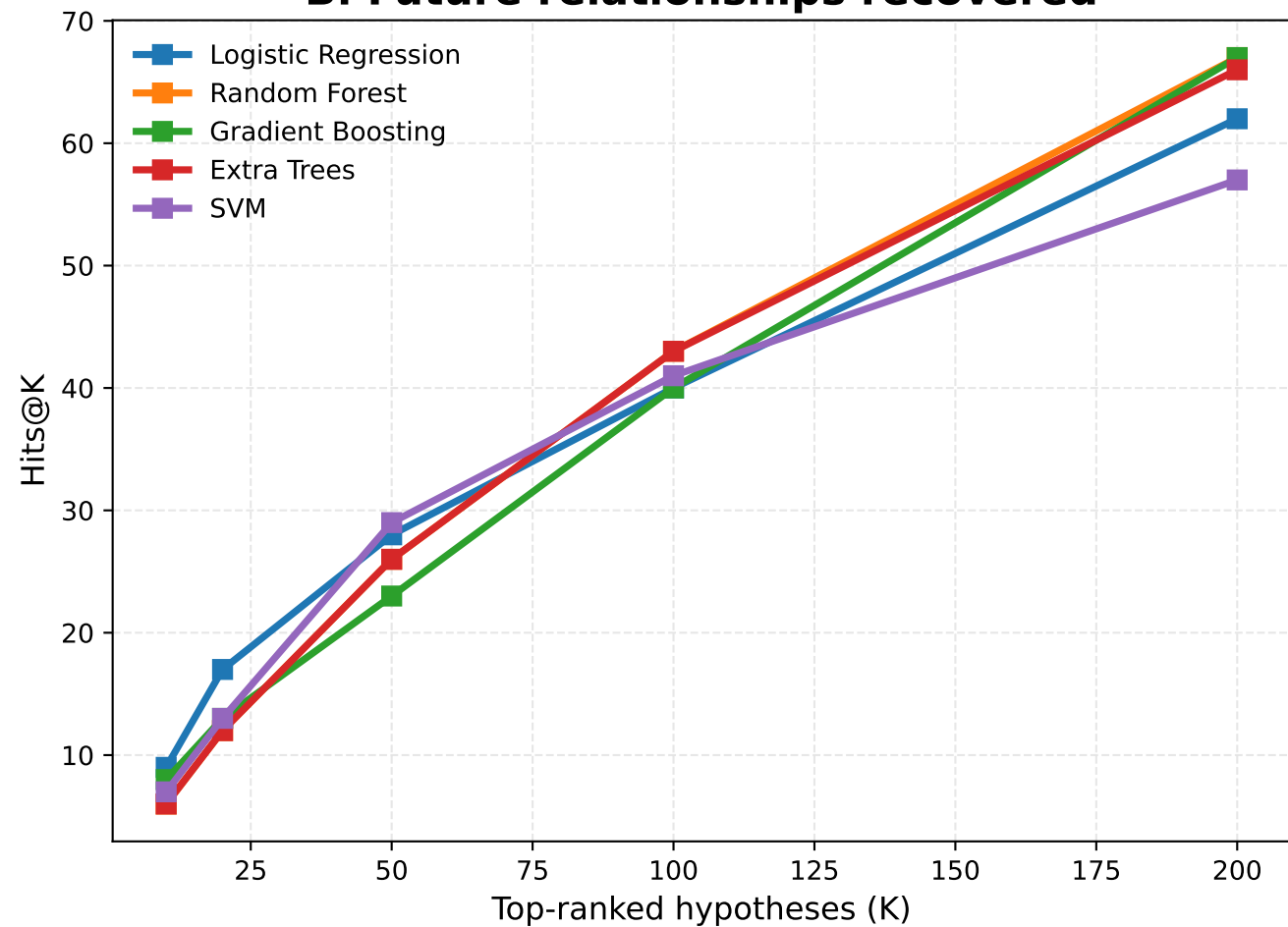


Strict global temporal validation of STX-LBD hypothesis ranking

A. Temporal Precision@K



B. Future relationships recovered



Performance was evaluated by testing pre-2016 AI-ranked dinoflagellate STX hypotheses against post-2015 literature.